

GOVERNANCE SIMULATION REPORT

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SCENARIO CONTEXT

A mid-level underwriting team processes life insurance applications near quarter-end under increasing operational pressure. Four decision scenarios are evaluated using an identical borderline-risk applicant profile with varying pressure and override conditions. The simulation identifies the precise checkpoint where admissibility breaks under operational stress.

SIMULATION SUMMARY

ID	Scenario	Pressure Level	Decision	Checkpoints
SCN-A	Baseline — No Pressure	None — Control	APPROVED	11/11
SCN-B	Moderate Pressure — Sales Team Pushing	Moderate — Sales push	BLOCKED	10/11
SCN-C	High Pressure — Escalation + Backlog	High — Escalation + Backlog	BLOCKED	7/11
SCN-D	Quarter-End Peak — Override Behavior A	Peak — Quarter-end override	BLOCKED	5/11

KEY FINDING — ADMISSIBILITY BREAKPOINT

Admissibility breaks at SCN-B — the first introduction of moderate sales pressure. CP-4 (Trend Persistence) detected underwriting condition instability (score 46.8 < threshold 50) before any fraud or AML signals were triggered. The engine blocked the decision based on structural instability in the underwriting environment, not on the applicant's individual profile.

SCENARIO DETAIL — CHECKPOINT RESULTS

SCN-A — Baseline — No Pressure

11/11 checkpoints passed

✓ APPROVED

Same borderline applicant. Standard review process. Full evidence documentation. No escalation pressure. Represents the control condition.

CP	Checkpoint Name	Signal	Score	Threshold	Result
CP-0	Signal Integrity Validation	signal_integrity	75.0	60	PASS
CP-1	Probability Check	probability_score	75.4	50	PASS
CP-2	Risk Limits	risk_exposure	59.0	65	PASS
CP-3	Signal Coherence	signal_coherence	79.7	55	PASS
CP-4	Trend Persistence	trend_persistence	52.8	50	PASS
CP-5	Stress Resilience	stress_resilience	75.7	35	PASS
CP-6	Logic Consistency	logic_consistency	64.5	40	PASS
CP-7	Temporal Coherence	temporal_coherence	65.0	45	PASS
CP-8	Edge Confirmation (ECW)	signal_coherence	79.7	48	PASS
CP-9	AML Gate	logic_consistency	64.5	38	PASS
CP-10	Fraud Detection Gate	signal_integrity	75.0	50	PASS

SCN-B — Moderate Pressure — Sales Team Pushing

10/11 checkpoints passed

X BLOCKED

Same applicant profile. Sales manager requesting approval to meet quarterly premium target. Evidence review partially expedited. Threshold interpretation relaxed informally.

CP	Checkpoint Name	Signal	Score	Threshold	Result
CP-0	Signal Integrity Validation	signal_integrity	75.0	60	PASS
CP-1	Probability Check	probability_score	62.0	50	PASS
CP-2	Risk Limits	risk_exposure	59.8	65	PASS
CP-3	Signal Coherence	signal_coherence	67.8	55	PASS
CP-4	Trend Persistence	trend_persistence	46.8	50	BLOCKED
CP-5	Stress Resilience	stress_resilience	70.1	35	PASS
CP-6	Logic Consistency	logic_consistency	47.0	40	PASS
CP-7	Temporal Coherence	temporal_coherence	65.0	45	PASS
CP-8	Edge Confirmation (ECW)	signal_coherence	67.8	48	PASS
CP-9	AML Gate	logic_consistency	47.0	38	PASS
CP-10	Fraud Detection Gate	signal_integrity	75.0	50	PASS

SCN-C — High Pressure — Escalation + Backlog

7/11 checkpoints passed

X BLOCKED

Same applicant profile. Underwriting backlog increasing. Senior sales manager escalation request filed. Evidence review incomplete. Exception review clause invoked. Policy-claim alignment degraded by override pressure.

CP	Checkpoint Name	Signal	Score	Threshold	Result
CP-0	Signal Integrity Validation	signal_integrity	75.0	60	PASS
CP-1	Probability Check	probability_score	46.6	50	BLOCKED
CP-2	Risk Limits	risk_exposure	61.1	65	PASS
CP-3	Signal Coherence	signal_coherence	56.4	55	PASS
CP-4	Trend Persistence	trend_persistence	38.4	50	BLOCKED
CP-5	Stress Resilience	stress_resilience	61.7	35	PASS
CP-6	Logic Consistency	logic_consistency	28.0	40	BLOCKED
CP-7	Temporal Coherence	temporal_coherence	65.0	45	PASS
CP-8	Edge Confirmation (ECW)	signal_coherence	56.4	48	PASS
CP-9	AML Gate	logic_consistency	28.0	38	BLOCKED
CP-10	Fraud Detection Gate	signal_integrity	75.0	50	PASS

SCN-D — Quarter-End Peak — Override Behavior Active

5/11 checkpoints passed

X BLOCKED

Same applicant profile. Quarter-end revenue pressure at peak. Multiple escalation overrides observed across the team. Evidence documentation minimal. Exception review invoked without documented justification. Override pattern flagged.

CP	Checkpoint Name	Signal	Score	Threshold	Result
CP-0	Signal Integrity Validation	signal_integrity	75.0	60	PASS
CP-1	Probability Check	probability_score	29.5	50	BLOCKED

CP-2	Risk Limits	risk_exposure	62.7	65	PASS
CP-3	Signal Coherence	signal_coherence	44.7	55	BLOCKED
CP-4	Trend Persistence	trend_persistence	28.8	50	BLOCKED
CP-5	Stress Resilience	stress_resilience	51.2	35	PASS
CP-6	Logic Consistency	logic_consistency	8.2	40	BLOCKED
CP-7	Temporal Coherence	temporal_coherence	65.0	45	PASS
CP-8	Edge Confirmation (ECW)	signal_coherence	44.7	48	BLOCKED
CP-9	AML Gate	logic_consistency	8.2	38	BLOCKED
CP-10	Fraud Detection Gate	signal_integrity	75.0	50	PASS

RECEIPT HASH CHAIN

Scenario	Content Hash	Prev Hash
SCN-A	e1ec408ff74c6d57	0000000000000000
SCN-B	626c74792bc436c8	e1ec408ff74c6d57
SCN-C	6c804cec975d9f3f	626c74792bc436c8
SCN-D	d0278f8fd452e8c6	6c804cec975d9f3f